An entropy chain rule on bounded Lipschitz domains

Kaiwen Jin
yui.ui.siki@gmail.com

October 12, 2026

Abstract

We prove an exact entropy chain rule for nonnegative densities satisfying a weak continuity equation with no boundary flux on any bounded Lipschitz domain. Finite kinetic action and Fisher information, together with finite initial entropy and weak continuity in time, imply absolute continuity of the entropy on the closed time interval. The formula holds for the prescribed flux, with vacuum allowed and without upper or lower density bounds. This extends the convex-domain chain rule to nonsmooth nonconvex domains. The proof reflects both the density and the flux across local boundary graphs. A truncated entropy identity then permits the localization terms to cancel before the logarithmic truncation is removed.

1 The chain rule

The entropy chain rule connects a weak continuity equation to the energy balance of a diffusion equation. For example, in the Fokker–Planck equation $\partial_t \rho = \operatorname{div}(\nabla\rho + \rho\nabla V)$ the flux is $F = -\nabla\rho - \rho\nabla V$. A chain rule for a general prescribed flux is also needed when identifying limits that satisfy an energy-dissipation inequality. Cancès, Monsaingeon and Natale prove such a rule on convex domains in [1, Proposition A.1]; they ask immediately afterwards for an extension to nonsmooth nonconvex Lipschitz domains.

Casteras, Flaim and Monsaingeon obtain an upper chain rule in smooth nonconvex geometry [2, Theorem 3]. Their exact formula for the optimal flux extends to a prescribed flux under an additional approximation condition on the logarithmic gradient [2, Theorem 4]; the concluding discussion of Section 4 of that version leaves Lipschitz geometry open. A related chain rule on Lipschitz domains is proved by Erbar and Meglioli [3, Proposition 5.15] for Dirichlet boundary data. Its dissipation functional incorporates the prescribed boundary trace [3, Definition 5.7]. Here the boundary condition is imposed only on the flux, in the weak formulation below.

Let $d \geq 1$, let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, and let $T > 0$. Put $Q = (0,T) \times \Omega$. For $V \in C^2(\overline{\Omega})$ and $\pi = e^{-V}$, define

$$E(\rho) = \int_\Omega (\beta(\rho) + \rho V)\, dx, \qquad \beta(r) = r\log r - r + 1, \qquad \beta(0) = 1. \tag{1}$$

The notation L^1_w means L^1 with its weak topology. All gradients are weak spatial gradients and all spacetime integrals are Lebesgue integrals. We set $|F|^2/\rho = 0$ where $(\rho, F) = (0,0)$ and $|F|^2/\rho = +\infty$ where $\rho = 0$ and $F \neq 0$.

Theorem 1.1. *Suppose that $\rho \geq 0$ belongs to $C([0,T]; L^1_w(\Omega))$, with $\int_\Omega \rho(t)\, dx = M > 0$ for every $t \in [0,T]$, and that $F \in L^1(Q;\mathbb{R}^d)$. Assume the no-flux weak continuity equation*

$$\int_0^T \int_\Omega (\rho\, \partial_t\varphi + F \cdot \nabla\varphi)\, dx\, dt = \int_\Omega \rho(T)\varphi(T)\, dx - \int_\Omega \rho(0)\varphi(0)\, dx \tag{2}$$

for every $\varphi \in C^1([0,T]\times\overline{\Omega})$*, and assume* $E(\rho(0)) < \infty$ *and*

$$\int_0^T \int_\Omega \left(\frac{|F|^2}{\rho} + 4\pi \left| \nabla \sqrt{\rho/\pi} \right|^2 \right) dx\, dt < \infty. \tag{3}$$

The Fisher term in (3) *requires* $\sqrt{\rho(t)/\pi} \in H^1(\Omega)$ *for almost every* t*. Then* $E(\rho(t))$ *is finite for every* $t \in [0,T]$*, is absolutely continuous on* $[0,T]$*, and satisfies, for all* $0 \le s \le t \le T$*,*

$$E(\rho(t)) - E(\rho(s)) = \int_s^t \int_\Omega \frac{F}{\sqrt{\rho}} \cdot \left(2\nabla\sqrt{\rho} + \sqrt{\rho}\, \nabla V \right) dx\, dt. \tag{4}$$

Here $F/\sqrt{\rho} = 0$ *on* $\{\rho = 0\}$*. In particular,*

$$\frac{d}{dt} E(\rho(t)) = \int_\Omega F(t) \cdot \nabla \log(\rho(t)/\pi)\, dx \quad \text{for almost every } t, \tag{5}$$

where the integrand means the pairing in (4)*.*

Thus the hypotheses and the prescribed-flux conclusion of [1, Proposition A.1] hold on every bounded Lipschitz domain, including those that are simultaneously nonsmooth and nonconvex. There is no density bound or smooth-gradient approximation hypothesis.

Spatial convolution proves the formula away from the boundary. At a Lipschitz graph, an even reflection of the density must be accompanied by a corresponding reflection of the flux. The resulting change of variables preserves the scalar entropy production even though it is not an orthogonal reflection. We first keep the entropy derivative bounded, so the terms created by a partition of unity are integrable and cancel. Only then do we recover the logarithm. Sections 2–3 give these two steps, and Section 4 proves the theorem.

2 A local truncated identity

We use elementary Sobolev composition facts in the following forms: a Sobolev gradient vanishes almost everywhere on a level set, and composition with a bi-Lipschitz map preserves $W^{1,p}$ with the almost-everywhere chain rule. These are [4, Corollary 21 and Theorem 31]; the absolute continuity on lines characterization in [4, Theorem 18] also justifies the product and reflection arguments below. Gradients here are column vectors, so for $z \circ S$ the composition formula is $\nabla(z \circ S) = DS^T (\nabla z) \circ S$.

Choose an even $\theta \in C^\infty(\mathbb{R}; [0,1])$ that equals one on $[-1,1]$, vanishes outside $[-2,2]$, and is nonincreasing in $|a|$. For $n \ge 1$, set

$$\kappa_n(r) = \theta(\log r/n) \quad (r > 0), \qquad \kappa_n(0) = 0, \qquad \beta_n''(r) = \frac{\kappa_n(r)}{r}, \qquad \beta_n(1) = \beta_n'(1) = 0. \tag{6}$$

We extend β_n'' by zero at $r = 0$. It vanishes near zero and for large r, so β_n is smooth on $[0,\infty)$ and β_n' is bounded for fixed n. Also $\beta_n \ge 0$.

Lemma 2.1. *Let* $U \subset \mathbb{R}^d$ *be bounded and open. Suppose that* $r \ge 0$ *belongs to* $C([0,T]; L^1_w(U))$*, with uniformly bounded mass,* $G \in L^1((0,T)\times U; \mathbb{R}^d)$*, and* $\partial_t r + \operatorname{div} G = 0$ *in distributions on* $(0,T)\times U$*. Assume* $\sqrt{r} \in L^2(0,T; H^1(U))$ *and*

$$\int_0^T \int_U \frac{|G|^2}{r}\, dx\, dt < \infty.$$

Use the same vacuum conventions as in Theorem 1.1. For every compactly supported Lipschitz function ψ on U and every $0 \le s \le t \le T$,

$$\begin{aligned}&\int_U \psi\beta_n(r(t))\,dx - \int_U \psi\beta_n(r(s))\,dx \\ &\quad= \int_s^t \int_U \left[\psi\kappa_n(r)\frac{G}{\sqrt{r}} \cdot 2\nabla\sqrt{r} + \beta_n'(r)G \cdot \nabla\psi\right] dx\,dt. \end{aligned} \tag{7}$$

No boundary condition on ∂U is required.

Proof. Put $v = \sqrt{r}$. The Sobolev product rule gives $r \in L^1(0,T;W^{1,1}(U))$ and $\nabla r = 2v\nabla v$. The gradient ∇v vanishes almost everywhere on $\{v = 0\}$. The action assumption also gives $G = 0$ there. Hence

$$a = G/v, \qquad b = 2\nabla v$$

are in $L^2((0,T)\times U)$, with $a = 0$ on vacuum.

Let η_ε be a nonnegative smooth spatial mollifier of unit integral, supported in the ε-ball. Extend r, G by zero solely to define $r_\varepsilon = \eta_\varepsilon * r$ and $G_\varepsilon = \eta_\varepsilon * G$. On a compact set $K \Subset U$, and for sufficiently small ε,

$$\nabla r_\varepsilon = \eta_\varepsilon * \nabla r, \qquad \partial_t r_\varepsilon = -\operatorname{div} G_\varepsilon.$$

The kernels in these identities stay inside U. Weighted Cauchy–Schwarz gives

$$\frac{|G_\varepsilon|^2}{r_\varepsilon} \le \eta_\varepsilon * |a|^2, \qquad \frac{|\nabla r_\varepsilon|^2}{r_\varepsilon} \le \eta_\varepsilon * |b|^2. \tag{8}$$

If $r_\varepsilon = 0$, both numerators vanish, and we set the normalized vectors equal to zero.

At almost every point where $r > 0$, spatial Lebesgue differentiation implies that these normalized vectors converge to a and b. At almost every vacuum point, $|a|^2 = |b|^2 = 0$, so (8) proves their convergence to zero as well. Spatial differentiation at almost every fixed time, followed by Fubini, gives the spacetime assertion. The convolutions of the zero-extended $|a|^2$ and $|b|^2$ converge in spacetime L^1 by continuity of spatial translations. They are therefore uniformly integrable. The bounds (8) and almost-everywhere convergence then imply

$$\frac{G_\varepsilon}{\sqrt{r_\varepsilon}} \longrightarrow a, \qquad \frac{\nabla r_\varepsilon}{\sqrt{r_\varepsilon}} \longrightarrow b \quad \text{strongly in } L^2((0,T)\times K). \tag{9}$$

Indeed, each squared difference is uniformly integrable and tends to zero almost everywhere, so its integral tends to zero.

For fixed ε, weak L^1 continuity of r implies continuity of $r_\varepsilon(t,x)$ in t against each convolution kernel. Uniform mass bounds give uniform spatial derivative bounds on K, hence continuity into $C^k(K)$ for each fixed k. Moreover, $\|\operatorname{div} G_\varepsilon(t)\|_{C^k(K)} \le C_{\varepsilon,k}\|G(t)\|_1$. Integrating the distributional equation in time therefore gives an absolutely continuous $C^k(K)$ representative, which agrees with the continuous one at every time. Applying the ordinary chain rule and integrating by parts in space yields

$$\begin{aligned}&\int_U \psi\beta_n(r_\varepsilon(t))\,dx - \int_U \psi\beta_n(r_\varepsilon(s))\,dx \\ &\quad= \int_s^t \int_U \left[\psi\beta_n''(r_\varepsilon)G_\varepsilon \cdot \nabla r_\varepsilon + \beta_n'(r_\varepsilon)G_\varepsilon \cdot \nabla\psi\right] dx\,dt. \end{aligned} \tag{10}$$

Take K to contain the support of ψ in its interior; integration by parts is valid for this Lipschitz cutoff.

At each fixed endpoint, $r_\varepsilon \to r$ locally in L^1, and β_n is Lipschitz. The endpoint terms thus converge for every s, t. In the first integrand of (10), write

$$\beta_n''(r_\varepsilon) G_\varepsilon \cdot \nabla r_\varepsilon = \kappa_n(r_\varepsilon) \frac{G_\varepsilon}{\sqrt{r_\varepsilon}} \cdot \frac{\nabla r_\varepsilon}{\sqrt{r_\varepsilon}}.$$

The products converge in L^1 by (9). The multiplier $\kappa_n(r_\varepsilon)$ is bounded and converges almost everywhere, including on vacuum. For the second integrand, $G_\varepsilon \to G$ in L^1, while β_n' is bounded and continuous; splitting off $G_\varepsilon - G$ and applying dominated convergence to the remaining term proves L^1 convergence. This gives (7). □

3 Reflection at the boundary

We first extend the no-flux identity (2) from C^1 tests on the closed domain to Lipschitz spatial tests. If ζ is bounded and Lipschitz on $\mathbb{R}^d$, smooth mollifications converge uniformly on $\overline{\Omega}$, their gradients are uniformly bounded, and those gradients converge almost everywhere in Ω. Dominated convergence against $F \in L^1$ extends (2) to $\varphi(t,x) = \alpha(t)\zeta(x)$, $\alpha \in C^1([0,T])$. Consequently

$$\int_\Omega \zeta \rho(t)\, dx - \int_\Omega \zeta \rho(s)\, dx = \int_s^t \int_\Omega F \cdot \nabla \zeta \, dx\, dt \quad (0 \le s \le t \le T). \tag{11}$$

The scalar curve on the left is continuous by weak L^1 continuity. Its distributional derivative is the displayed L^1 function, so its absolutely continuous representative agrees at every time. A bounded Lipschitz function on $\overline{\Omega}$ can be extended to $\mathbb{R}^d$ and treated in the same way.

Proposition 3.1. *Suppose that ρ, F have the continuity and no-flux properties of Theorem 1.1, and that $u = \sqrt{\rho} \in L^2(0,T; H^1(\Omega))$ and $F/u \in L^2(Q; \mathbb{R}^d)$. Then, for every $n \ge 1$ and every $0 \le s \le t \le T$,*

$$\int_\Omega \beta_n(\rho(t))\, dx - \int_\Omega \beta_n(\rho(s))\, dx = \int_s^t \int_\Omega \kappa_n(\rho) \frac{F}{u} \cdot 2\nabla u \, dx\, dt. \tag{12}$$

Proof. Near a boundary point, take rigid coordinates in which Ω lies above a Lipschitz graph $x_d = g(x')$. By shrinking the chart, choose a ball $B \subset \mathbb{R}^{d-1}$ and $h > 0$ so that

$$U = \{(x', x_d) \in B \times \mathbb{R} : \ |x_d - g(x')| < h\},$$
$$U_+ = U \cap \Omega = \{(x', x_d) \in B \times \mathbb{R} : \ 0 < x_d - g(x') < h\}$$

lie within the graph neighborhood. Define

$$R(x', x_d) = (x', 2g(x') - x_d), \qquad A(x) = DR(x) = \begin{pmatrix} I_{d-1} & 0 \\ 2\nabla g(x')^T & -1 \end{pmatrix} \quad \text{almost everywhere.} \tag{13}$$

The map R is a bi-Lipschitz involution, $A(Rx) = A(x)$ and $A(x)^2 = I$. It preserves volume: this follows by the one-dimensional substitution in x_d for each x' and Fubini. Its derivative is bounded in terms of $\mathrm{Lip}(g)$. In dimension one, use ordinary reflection about an endpoint.

Extend the actual density and flux to U by

$$r(x) = \rho(x), \quad G(x) = F(x) \quad (x \in U_+), \qquad r(Rx) = \rho(x), \quad G(Rx) = A(x)F(x) \quad (x \in U_+). \tag{14}$$

The density extension is a bounded linear map on L^1, so it preserves weak continuity in time and gives a uniform mass bound. To check $\sqrt{r} \in L^2(0,T; H^1(U))$, flatten the graph by $S(y', y_d) =$

$(y', y_d + g(y'))$. Composition with S maps $u|_{U_+}$ to $H^1(B \times (0,h))$. Even reflection across $y_d = 0$ belongs to $H^1(B \times (-h,h))$: on almost every vertical line the one-dimensional H^1 representative has a limit at zero and its even extension is absolutely continuous; the tangential derivatives are reflected unchanged. The absolute continuity on lines characterization then applies. Composing back with S^{-1} gives $\sqrt{r}$ and the gradient identity

$$\nabla\sqrt{r}(Rx) = A(x)^T \nabla u(x) \quad \text{for almost every } x \in U_+. \tag{15}$$

These composition facts also give the required time-integrated norm bound. The reflected kinetic action is bounded because $G(Rx)/\sqrt{r(Rx)} = A(x)F(x)/u(x)$. Only first derivatives of the graph occur.

For $\phi \in C_c^\infty(U)$, the folded spatial test $\zeta(x) = \phi(x) + \phi(Rx)$ on U_+, extended by zero on the rest of Ω, is Lipschitz on $\overline{\Omega}$ and has an ambient Lipschitz extension. Its support stays away from the outer chart boundary. Using (11) and changing variables on the lower half of U proves $\partial_t r + \operatorname{div} G = 0$ in $(0,T) \times U$, since

$$\nabla\zeta(x) = \nabla\phi(x) + A(x)^T \nabla\phi(Rx).$$

Thus Lemma 2.1 applies to the reflected pair. No normal trace of F is needed.

Choose a finite smooth nonnegative ambient partition of unity χ_j, whose sum is one on a neighborhood of $\overline{\Omega}$. Each support is compactly contained in an interior patch or a boundary chart U of the preceding kind. Such a partition follows from a finite open cover of the compact closure by choosing smaller neighborhoods and smooth bumps. In an interior patch apply (7) with $\psi = \chi_j$, $r = \rho$, $G = F$. For a boundary chart use the cutoff

$$\psi(x) = \chi_j(x) \quad (x \in U_+), \qquad \psi(Rx) = \chi_j(x) \quad (x \in U_+).$$

It is compactly supported and Lipschitz on U, as follows by composition with $(x', x_d) \mapsto (x', g(x') + |x_d - g(x')|)$, and $\nabla\psi(Rx) = A(x)^T \nabla\chi_j(x)$. Both pairings in (7) are invariant under this reflection:

$$\begin{aligned} (AF/u) \cdot (2A^T \nabla u) &= (F/u) \cdot 2\nabla u, \\ (AF) \cdot (A^T \nabla\chi_j) &= F \cdot \nabla\chi_j. \end{aligned} \tag{16}$$

Indeed, $A^2 = I$, although A need not be orthogonal. All three terms in (7) are therefore twice their upper-half integrals. Dividing by two gives the same formula as in an interior patch:

$$\begin{aligned} &\int_\Omega \chi_j \beta_n(\rho(t))\,dx - \int_\Omega \chi_j \beta_n(\rho(s))\,dx \\ &\quad = \int_s^t \int_\Omega \left[\chi_j \kappa_n(\rho)(F/u) \cdot 2\nabla u + \beta_n'(\rho) F \cdot \nabla\chi_j\right] dx\,dt. \end{aligned}$$

Summing over j cancels the last terms because $\sum_j \nabla\chi_j = 0$ on Ω. This proves (12). The cancellation takes place for fixed n, when $\beta_n'(\rho)F$ is integrable; it requires no integrability of $\log(\rho)F$. □

4 Removal of the truncation

Proof of Theorem 1.1. Recall $\pi = e^{-V}$, with $V \in C^2(\overline{\Omega})$. Set $w = \sqrt{\rho/\pi}$ and $u = \sqrt{\rho} = \sqrt{\pi}\, w$. Since $\sqrt{\pi}$ and its reciprocal are bounded smooth multipliers,

$$2\nabla u + u\nabla V = 2\sqrt{\pi}\,\nabla w, \qquad 4\int_Q |\nabla u|^2\,dx\,dt \le 8\int_Q \pi|\nabla w|^2\,dx\,dt + 2TM\|\nabla V\|_\infty^2. \tag{17}$$

Also $\int_Q u^2\,dx\,dt = TM$. Thus $u \in L^2(0,T;H^1(\Omega))$, and (3) gives $F/u \in L^2(Q)$, with $F = 0$ on vacuum. Proposition 3.1 applies.

The truncations increase to β. More explicitly,

$$\beta_n(r) = \begin{cases} \displaystyle\int_1^r (r-a)\frac{\kappa_n(a)}{a}\,da, & r \geq 1, \\ \displaystyle\int_r^1 (a-r)\frac{\kappa_n(a)}{a}\,da, & 0 \leq r \leq 1. \end{cases}$$

The kernels are nonnegative and $\kappa_n(a) \uparrow 1$ for $a > 0$. Monotone convergence gives $\beta_n(r) \uparrow \beta(r)$, including $\beta_n(0) \uparrow 1$. The right-hand side of (12) is dominated by $|F/u|\,2|\nabla u| \in L^1(Q)$ and tends to the untruncated pairing. Since V is bounded and the mass is finite, $\int_\Omega \beta(\rho(0))\,dx < \infty$ follows from finite initial E. Taking $s = 0$ in (12) first, monotone convergence on the endpoints and dominated convergence on the right prove finiteness at every time and

$$\int_\Omega \beta(\rho(t))\,dx - \int_\Omega \beta(\rho(0))\,dx = \int_0^t \int_\Omega (F/u)\cdot 2\nabla u\,dx\,dt.$$

Subtracting two such identities gives the formula for every s, t, including both endpoints of the actual weakly continuous curve.

Finally (2) with spatial test V gives

$$\int_\Omega V\rho(t)\,dx - \int_\Omega V\rho(s)\,dx = \int_s^t \int_\Omega F\cdot\nabla V\,dx\,dt.$$

Adding the two formulas proves (4). By (17) its integrand is $(F/\sqrt{\rho})\cdot 2\sqrt{\pi}\,\nabla\sqrt{\rho/\pi}$, the product of the two L^2 factors in (3). It belongs to $L^1(Q)$. Hence (4) proves absolute continuity and differentiation gives (5) almost everywhere. □

AI use. The proof development, source searches, drafting, and verification of this note used OpenAI Codex. Automated reviews are not independent mathematical certification. The manuscript is submitted for independent mathematical review.

References

[1] C. Cancès, L. Monsaingeon and A. Natale, *Discretizing the Fokker–Planck equation with second-order accuracy: a dissipation driven approach*, Numerische Mathematik **158** (2026), 1513–1563. doi:10.1007/s00211-026-01537-3. Preprint: arXiv:2410.03367v1. The cited proposition numbering refers to this preprint version.

[2] J.-B. Casteras, M. Flaim and L. Monsaingeon, *Entropy and Fisher information in non-convex domains: one chain to rule them all*, arXiv:2512.05826v2, revised September 16, 2026.

[3] M. Erbar and G. Meglioli, *Gradient flow for a class of diffusion equations with Dirichlet boundary data*, arXiv:2408.05987v2, February 27, 2025.

[4] P. Hajłasz, *Sobolev spaces, theory and applications*, lecture notes, Jyväskylä Summer School, 1998. https://sites.pitt.edu/~hajlasz/Notatki/jyv-98c.pdf.